

Bachelor's Programme in Information Networks

Validity and long-term stability of sleep metrics in consumer wearable devices

Examining the performance, transparency, and end-user implications

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Bachelor's thesis
2026

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Title of thesis Validity and long-term stability of sleep metrics in consumer wearable devices		
Programme Bachelor of Science in Technology		
Major Information Networks		
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Date 15.05.2026	Number of pages 49	Language English

Abstract

Consumer wearable devices, such as smartwatches and rings, are increasingly used to measure sleep in everyday life. Wearables offer an opportunity to scale current sleep measurement practices and to measure sleep longitudinally in naturalistic environments. However, their accuracy and reliability must be evaluated, especially in long-term use. Evaluation is difficult because devices function as proprietary black boxes, leaving the internal logic hidden from the user. Rapid technological evolution further complicates examining results and comparability over time. This thesis examined the validity and longitudinal stability of sleep metrics produced by consumer wearable devices. It was conducted as a literature review focusing on performance evaluation of sleep metrics, transparency, longitudinal comparability, and end-user implications.

The review shows that the reported accuracy of sleep metrics is affected by both the device and methodological factors. Wearables can provide approximate estimates of sleep, such as wake and sleep detection, but accuracy remains metric and device specific and varies across the population and setting studied. The longitudinal stability of metrics is complicated by the black-box nature of the system, as it is difficult to determine when changes in results reflect changes in the user's sleep, rather than in the underlying device or algorithm.

The thesis suggests that wearable sleep measurement should be evaluated not only through device accuracy, but also whether measurements remain stable, comparable, and interpretable over time. This is essential as longitudinal use is one of the key advantages of wearable sleep measurement, yet this is where limited transparency complicates comparability and interpretability of results.

Keywords wearables, sleep tracking, performance evaluation, algorithm transparency, standardisation, longitudinal monitoring

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Datum 15.05.2026	Sidantal 49	Språk engelska

Sammandrag

Användningen av bärbara konsumentenheter, så som smartklockor och ringar, har blivit allt vanligare för att mäta sömn i vardagen. Bärbara enheter möjliggör att skala nuvarande sömnforskningsmetoder och mäta sömn longitudinellt och i naturliga förhållanden. Samtidigt behöver deras noggrannhet och tillförlitlighet utvärderas särskilt inom longitudinell användning. Detta är en utmaning, eftersom enheterna är proprietära och systemen gömmer den interna logiken från användaren. Dessutom försvårar den snabba teknologiska utvecklingen utvärderingen av enheterna över tid på grund av att nya enheter och versioner introduceras regelbundet på marknaden. Följaktligen är den långsiktiga tillförlitligheten hos mätningarna fortfarande osäker. Denna avhandling undersöker validiteten och långtidsstabiliteten av sömndata från bärbara konsumentenheter. Avhandlingen genomfördes som en litteraturgranskning och fokuserar på utvärdering av sömnmått, transparens, longitudinell jämförbarhet och konsekvenser för slutanvändare.

Avhandlingen visar att den rapporterade noggrannheten hos sömnmått påverkas både av själva enheten och metodologiska faktorer. Bärbara enheter kan ge riktningsgivande uppskattningar av sömn för detektering av vakenhet och sömn, men noggrannheten är beroende av måttet och enheten, samt på populationen och miljön där mätningen utförs. Den longitudinella stabiliteten kompliceras av den begränsade transparensen, eftersom det är svårt att avgöra om förändringarna i sömnmåtten speglar förändringar i användarens sömn eller i den underliggande teknologin.

Arbetet föreslår att sömnmätning med bärbara enheter utvärderas utifrån mätresultatens stabilitet, jämförbarhet och tolkbarhet, istället för att enbart fokusera på noggrannhet i jämförelse med polysomnografi. Detta är viktigt, eftersom långtidsanvändning är en av de främsta fördelarna med bärbara enheter, men vid longitudinell användning försvårar den begränsade transparensen tolkningen av sömnmått i störst utsträckning.

Nyckelord bärbara enheter, sömnmätning, utvärdering, algoritmisk transparens, standardisering, longitudinell mätning

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Preface and acknowledgements

I want to thank my thesis advisor, Dr Talayeh Aledavood, for the advice and guidance throughout the process. I would also like to thank my friends and family for their feedback and encouragement.

Otaniemi, 15 May 2026
Karla Itkonen

Abbreviations

AASM	American Academy of Sleep Medicine
CWD	Consumer wearable devices
CWT	Consumer wearable technologies
EEG	Electroencephalography
NREM	Non-rapid eye movement
PPG	Photoplethysmography
PSG	Polysomnography
REM	Rapid Eye Movement
TST	Total sleep time

1 Introduction

Sleep research has expanded considerably in recent years, alongside the growing use of wearable devices to track and analyse sleep patterns among consumers (T. Lee et al., 2023; Seçkin et al., 2023). This growth is also reflected in the commercial market, which is projected to reach 58.21 billion dollars by 2030 (Chapman, 2026). At the same time, technological advancements have increased both the capabilities and accessibility of these devices. As consumer adoption rises, ensuring the validity and reliability of the sleep data they produce has become increasingly essential.

The rapid pace of technological advancements has raised concerns about the validity of data collected from consumer wearable technologies (CWT). This has led to an increase in research validating sleep data, as it is both relevant for ensuring data reliability and examining remaining challenges in data accuracy (T. Lee et al., 2023). Reliable technology plays a key role in understanding sleep patterns and their health implications and is essential to ensure comparability and transparency within this field of research (Birrer et al., 2024).

This study aims to understand the validity and stability of sleep metrics produced by consumer wearables over time as algorithms and hardware evolve. The goals are to examine how these changes affect the reliability of sleep data over time and the resulting outcomes for end users. This is done through answering the following three research questions:

RQ1. How are sleep metrics from consumer wearables evaluated, and how accurately do they measure sleep?

RQ2. To what extent are changes in algorithms and hardware in wearable devices considered in the validation and long-term comparability of sleep metrics?

RQ3. What are the implications for end users using consumer wearable technologies as accuracy evolves and transparency is limited?

By answering these questions, the thesis examines how sleep data is evaluated, its broader implications, and future needs. The thesis aims to provide a novel perspective on the evolving validity of sleep metrics, as current work mostly focuses on short-term validation. The study is a literature review based on academic and peer-reviewed sources. It aims to narrow gaps in the literature concerning the validity, comparability and transparency of consumer wearable technologies used to measure sleep.

The thesis examines consumer wearable technologies for sleep measurement, focusing on the evaluation of measurements, system transparency, technological change over time, and the resulting implications for end users. The rest of this thesis is organised as follows. Chapter 2 presents the methodology and scope of the review. Chapter 3 introduces key concepts and terminology for consumer wearable sleep measurement and sleep evaluation. Chapter 4 reviews the literature on performance evaluation, transparency challenges, continuity, and outlines the implications for end users. Lastly, in chapter 5, the thesis's findings and implications are discussed, including directions for future research and the work's reliability.

2 Methodology and scope

This thesis consists of a literature review examining how sleep metrics from consumer wearables are evaluated and how issues related to transparency, accuracy, and long-term stability are discussed. Furthermore, the thesis examines the long-term implications for end users, as there is limited transparency relating to how consumer wearable devices evolve.

The review focuses on articles published between 2018 and 2026. Articles published before 2018 were included only if they acted as the original source in a selected article. This range was chosen to ensure that the literature reflects recent advances in consumer wearable technologies. Recent studies are particularly relevant because the core research question explores the evolution of the technologies used, their accuracy as tools for sleep measurement, and the implications for users. As devices evolve rapidly with new generations and software releases, older papers have been excluded to keep the review aligned with the current technological landscape and methodological challenges.

The review primarily includes peer-reviewed articles that are published in English. One non-peer-reviewed source by Oura (n.d.) provides a perspective on device update frequency, as no peer-reviewed sources were identified that document consumer wearable updates with sufficient details.

Literature searches were conducted through Google Scholar, Scopus, PubMed, Oxford Academic, and ACM Digital Library. Search terms used in the databases can be grouped into four themes: sleep measurement and device (metric, wearable technology, wearable sleep sensor, sleep sensing, sleep architecture, evaluation); underlying technology (algorithm, transparency, black box, proprietary algorithm); longitudinal stability and technological advancement (consistency, evolution, advancement, long-term, longitudinal, stability, drift, agreement, repeatability); and user perspective (user, perception, and implications). The searches were conducted by using Boolean operators to connect the search terms to enable both individual searches and combinations of search terms. An example of a used combination is: ('consumer wearable' OR 'wearable' or smartwatch or Oura) AND (sleep or 'sleep measurement' OR 'sleep metric' OR 'sleep data') AND (longitudinal OR stability OR agreement OR 'over time' OR repeatability OR 'measurement drift'). Additional articles were derived from the bibliographies of identified articles.

Articles were selected by screening the title, abstract and full text. Articles were deemed relevant and included in the review if they addressed wearables for sleep measurement, performance evaluation of sleep metrics produced by

wearables, transparency of devices and their outputs, consumer wearables, comparison of wearables against clinical standards, the technological comparability of devices, and implications for users using wearables long-term. Wrist-worn devices and smart rings were prioritised over headbands and chest belts, as these better represent devices commonly used by consumers.

Literature on sleep measurement that did not relate measurement to wearables or consumer wearables was excluded. Measurements using only mobile applications or bed sensors are excluded from this review. In addition, articles where wearables were not associated with sleep monitoring (e.g. used for fitness tracking) or did not include wearables as the primary measurement tool are outside of the scope of this review. Furthermore, this review does not distinguish between clinical and consumer-grade wearable devices as an inclusion factor. Studies using wearable devices to evaluate sleep metrics were included regardless of their status as clinically approved or not. Studies with the primary purpose of diagnosing clinical sleep conditions or other disorders through wearables or gold standard measurement were excluded.

AI has been used in the screening process for summarising article content, but all final decisions were made by the author. The used model includes OpenAI's ChatGPT versions GPT-5.3 and GPT-5.4.

3 Background

This chapter explains the core definitions, concepts, and methods used in sleep research. It establishes the foundation for what consumer wearables are, what they measure, and their role in sleep research. The first section, 3.1, outlines the structure of sleep and the measures used in clinical sleep research. Next, section 3.2 describes the gold standard in sleep research, polysomnography. Finally, section 3.3 describes the field of wearable devices, consumer wearables and wearable technology.

3.1 Sleep architecture and clinical sleep measures

Sleep can broadly be divided into three main phases: light, deep, and rapid eye movement (REM) (Pan et al., 2020). However, Pan et al. (2020) note that sleep involves cyclical transitions between stages; thus, the stages should not be considered separate entities. Throughout the night, sleep progresses as repeating cycles lasting approximately 90 minutes (Rentz et al., 2021). The structure of sleep stages within cycles changes as sleep progresses (Rentz et al., 2021). Commonly, light sleep transitions to deep sleep, and then to REM sleep.

Sleep architecture refers to the structure of sleep, comprising its distinct stages. A common standard followed in classifying sleep stages is the American Academy of Sleep Medicine (AASM) scoring manual (Pan et al., 2020). This standard recognises five stages of sleep: wake, stage N1, stage N2, stage N3, and stage R (REM sleep). Stages N1 and N2 are light sleep, while N3 is deep sleep (Imtiaz, 2021). According to Pan et al. (2020), some studies use a simpler classification of sleep stages, whereas others include an additional fourth stage of deep sleep (N4). The fourth stage stems from the earlier Rechtschaffen and Kales (R&K) classification, which divides non-rapid eye movement sleep (NREM) into four stages instead of three (Imtiaz, 2021). The currently used AASM classification standard is based on the R&K standard (Van Gorp et al., 2022).

In clinical research, sleep quality is assessed using the standard parameters defined by AASM, such as sleep efficiency, total sleep time (TST), sleep latency, sleep stages N1 and N2, REM sleep, wake time after sleep onset, and nocturnal time (Pan et al., 2020). These parameters are widely accepted and serve as the reference point in research. Subsequently, consumer device metrics are often interpreted in relation to these clinical measures, as there are no existing standards for analysing sleep stages using only signals collected from wearables (Rentz et al., 2021).

3.2 Polysomnography

Polysomnography (PSG) is a diagnostic tool used to evaluate sleep and is widely considered the gold standard in sleep evaluation (Pan et al., 2020). This is because PSG provides a comprehensive collection of physiological signals during sleep (Birrer et al., 2024; Djanian et al., 2022). PSG includes electroencephalography (EEG), which measures brain activity, and is highly informative when classifying sleep stages (Zapalac et al., 2024).

Polysomnography combines four physiological measurement methods, including Electroencephalography, Electrooculography, Electromyography and Electrocardiography (Djanian et al., 2022). In practice, multiple sensors are attached to the body and measure physiological signals related to brain activity, cardiac activity, muscle contractions, eye movements, breathing, and oxygen saturation (Birrer et al., 2024; Mathunjwa et al., 2025). Following data collection, the data are analysed in epochs, which are 30-s segments of sleep, and then classified into sleep stages (Birrer et al., 2024). Assigning a sleep stage to a specific epoch is known as sleep staging or sleep scoring (Imtiaz, 2021). Classification is performed manually by clinicians according to the AASM guidelines (Kazemi et al., 2024). Classification can be completed by multiple clinicians simultaneously, where each clinician's results are compared. The agreement between the classifications of the clinicians' results is referred to as inter-rater agreement (Van Gorp et al., 2022).

Combining analysis of sleep stages with signs of other sleep disturbances, such as snoring or breathing pauses, enables the identification and diagnosis of sleep disorders. Although PSG is considered the gold standard for accurately measuring sleep, it is primarily used in clinical settings (Mogavero et al., 2025). Its widespread use is limited by the intrusive nature of measurement and high cost, which in turn affects the number of nights recorded (Zapalac et al., 2024). For this reason, PSG provides a strong reference for accurate sleep evaluation, but is less suitable in naturalistic environments and long-term settings. These limitations are why consumer wearables have become increasingly relevant for individual sleep monitoring and for sleep research.

3.3 Sleep monitoring with wearable devices and underlying technology

Wearable devices are non-invasive sensors worn on the body that collect and analyse physiological signals, such as heart rate and sleep patterns (Seçkin et al., 2023). Óskarsdóttir et al. (2022) describe wearables as the umbrella term for sensors worn on the body. Consumer-grade wearables consist of devices aimed at personal health and activity monitoring (Óskarsdóttir et al., 2022). They enable continuous measurement of physiological signals in the user's natural setting, which offers advantages over clinical measurement methods (Godfrey et al., 2018). Common devices include wrist-worn smartwatches and rings. Their main purpose is often to inform consumers about personal wellness; however, this poses challenges for the use of the devices in clinical settings and for how they should be evaluated in research. The core challenge is the proprietary nature of the underlying technology used in consumer wearables (Mogavero et al., 2025). Mogavero et al. (2025) recognise that consumer wearables are associated with numerous factors that affect their accuracy and reliability. These include undisclosed proprietary algorithms, sensor limitations, limited raw data access, and rapid technological advancement.

Consumer wearable technology refers to the underlying sensing technologies, processing techniques, and algorithms used in consumer wearable devices (CWD). The technology is embedded in CWDs, and the device only represents the user interface layer. Many consumer wearables come with companion applications that allow data visualisation (Mathunjwa et al., 2025). Additionally, they integrate with broader health ecosystems, thereby enabling data sharing.

Consumer wearable devices incorporate a variety of sensors that collect physiological data. The common sensors used in the devices are accelerometers, which measure movement, and photoplethysmography (PPG) sensors, which measure pulse and pulse variability (Birrer et al., 2024; Ryals et al., 2023). Often, these devices also integrate additional sensors, such as temperature and skin conductance sensors, to collect more varied data and provide a more comprehensive health picture to the consumer (Imtiaz, 2021). According to Godfrey et al. (2018), most devices rely on similar combinations of sensors and therefore collect largely the same types of data, which leads to differentiation between devices stemming from the computational methods and visualisation tools utilised to process and visualise the data.

Wearable devices provide consumers with a range of metrics describing sleep characteristics, including TST, sleep efficiency, and sleep structure breakdown (Berry et al., 2020). While some of the provided metrics align with clinically recognised measures, others are device-specific and lack clinical equivalents. As the produced outputs are derived from underlying

sensing technologies and processed by undisclosed algorithms, the results can represent different things depending on the device (Mogavero et al., 2025).

It is important to distinguish that consumer wearables do not collect the same physiological signals on which the AASM is based. Whereas AASM scoring is based on polysomnography data, consumer wearables rely on alternative, secondary physiological signals and do not measure sleep directly through the same signals as the gold standard (Rentz et al., 2021). Instead, CWDs provide sleep parameters as aggregate sleep indicators and should be understood as proxies for sleep (de Zambotti et al., 2024).

4 Literature review

This chapter reviews the literature on consumer wearable sleep technologies through four perspectives. First, it examines how wearable-derived sleep metrics are evaluated (4.1), focusing on performance evaluation and validation practices. Second, it considers the transparency of underlying systems and the interpretability of device outputs (4.2). Third, the literature is analysed from a longitudinal perspective, focusing on technological advancements, the stability of results, and comparability over time (4.3). Lastly, the implications for consumers using consumer wearables for sleep monitoring are outlined (4.4). The chapter ends with a summary table of the key findings (4.5).

4.1 Performance evaluation of wearable sleep metrics

The practical value of consumer wearables is dependent on whether the sleep metrics they provide are accurate. As adoption of these devices grows among both consumers and researchers, systematic evaluation against a reliable reference standard is essential to determine in which settings wearables can be reliably utilised. The following section first examines the performance evaluation of wearable devices' sleep metrics and their agreement with clinical standards. Thereafter, it examines the methodological factors shaping reported performance. Lastly, ending with approaches used to improve accuracy.

Reported performance

Performance evaluation of wearable devices is performed by comparing sleep results from a wearable device with a reference method (Menghini et al., 2021). The output is then compared based on time intervals consisting of sleep epochs, providing information about the device's performance accuracy. For consumer wearables, the reference is most frequently done against polysomnography (PSG) (Baron et al., 2018). Alternative methods of reference are EEG or actigraphy, but these are not as widely used (Baron et al., 2018).

Reported performance in wearables is commonly noted by terms such as sleep sensitivity, sleep specificity and sleep accuracy. Sleep sensitivity refers to the ability of a device to correctly recognise sleep, whereas sleep specificity corresponds to the ability to correctly recognise wakefulness (Ghorbani et al., 2022). Sleep accuracy, in turn, refers to the proportion of epochs classified

correctly when comparing wearable output with the reference scoring (Ghorbani et al., 2022).

The literature indicates that CWDs exhibit variable performance in sleep estimation. Y. J. Lee et al. (2025) conducted a meta-analysis of 24 studies comparing consumer wrist-worn wearable devices to PSG. The authors found significant differences in key sleep metrics in comparison to PSG, including TST, sleep efficiency, sleep latency, and wake after sleep onset. Accuracy of sleep metrics varied across studies and depended on the device used, suggesting that wearables are not yet clinically comparable to the gold standard. Guillodo et al. (2020) similarly note that variability in wearable measures and unreliable measurement accuracy present limitations to applicability in clinical settings.

A consistent finding across evaluation studies is that wearables tend to overestimate TST (Baron et al., 2018; Doherty et al., 2024; Guillodo et al., 2020; Mahir et al., 2025). This bias stems from low sensitivity to stages of wakefulness (Baron et al., 2018). Kainec et al. (2024) add to this by identifying proportional bias in sleep measures across five consumer sleep-tracking devices, demonstrating that wearables overestimate short periods of wakefulness, while long wake periods are underestimated.

Difficulties sensing wakefulness persist even as wearables' overall performance increases. De Zambotti et al. (2024) and T. Lee et al. (2023) argue that the sensing accuracy compared to the gold standard has improved, and in some cases reached high agreement. However, de Zambotti et al. (2024) highlight that this increase in agreement can be limited to specific sleep metrics. While sensitivity for sleep is often greater than 90%, wake specificity ranges from 20% to 70% (Chinoy et al., 2022; Lujan et al., 2021). This indicates that improvements in wearable performance remain metric-specific, with performance increasing for sleep detection, while wakefulness detection remains a challenge.

Sleep stage detection remains the weakest area of wearable performance. Detection of sleep stages is a valuable clinical property, as accurate sleep stage detection functions as the foundation for sleep disorder treatment and recognition (Birrer et al., 2024). However, the accuracy of wearables' sleep stage detection remains inconsistent across studies and devices (Mathunjwa et al., 2025). Mathunjwa et al. (2025) argue that wearables struggle, especially with recognising REM, deep, and light sleep. Birrer et al. (2024) found that the average sleep staging accuracy for two stages across the 53 devices assessed in the review was 87.2%. However, accuracy dropped to 69.7% for three-stage detection and to 65.2% for four-stage detection. As also

noted by Chinoy et al. (2022), wearables in their current form should therefore be used for wake and sleep detection rather than stage detection.

In aggregate, wearables perform reasonably well when constrained to only sleep and wake detection, reaching approximately 80-90 % agreement with PSG. However, wearables exhibit systematic biases toward overestimating sleep, difficulty detecting wakefulness, and poor precision in recognising sleep stages. Consequently, while wearables provide an approximate picture compared to PSG, their results do not yet meet the accuracy required by clinical standards.

Methodological limitations

Before considering how wearable performance has improved, it is worth examining how methodological choices themselves shape the evaluation of device performance. Mogavero et al. (2025) identify four methodological considerations when evaluating consumer wearables: sensor fidelity, algorithm transparency, population variability, and environmental influence on results. Together with the evaluation method and the use of PSG as a reference standard, these factors help explain why the reported performance of wearable-device sleep metrics varies across studies. These factors are illustrated in Figure 1.

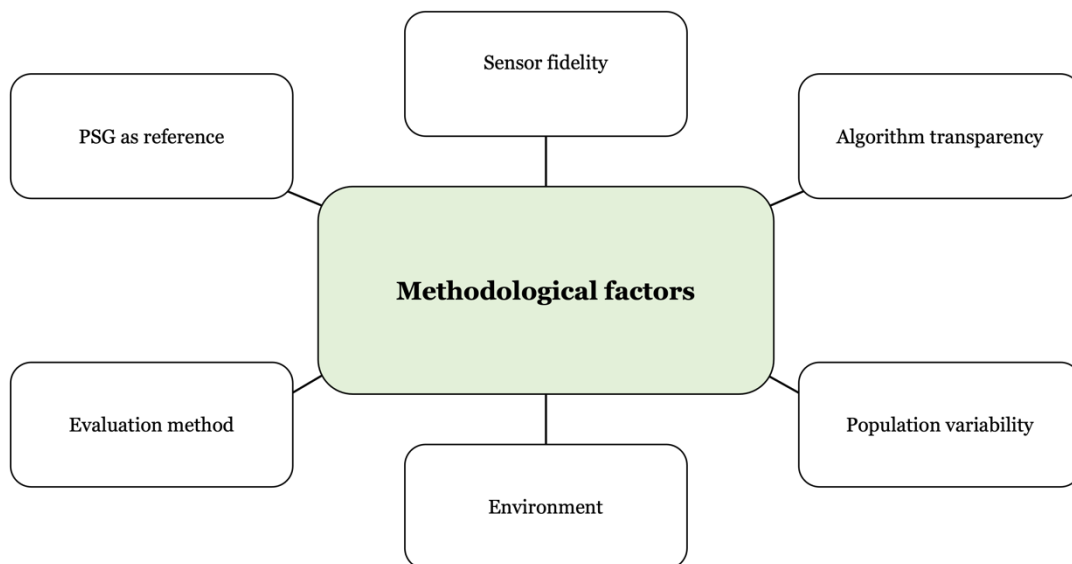


Figure 1: Methodological factors affecting wearable performance.

Sensor fidelity can be understood more broadly as a set of device properties that enable precise data collection. This includes the resolution of the sensors embedded in wearables, the physiological signals they can capture, and the

precision of signal measurement (Mogavero et al., 2025). In this sense, the accuracy of sleep metrics depends on the sensing technology used. As noted by Imtiaz (2021), wearable sleep monitoring faces a trade-off between usability and accuracy. This means that the level of accuracy that wearable-produced sleep metrics can achieve depends on the chosen modality and the sensing capabilities of the device and its sensors (Mogavero et al., 2025). Rentz et al. (2021) effectively capture this idea by arguing that algorithm accuracy is driven by the quality of the underlying signal. Thereby highlighting that the output reflects the input accuracy. This means that the sensing features used in devices determine the accuracy a device can achieve, which can result in variability in measurements depending on the device examined.

A second source of variability stems from algorithm transparency. Even when devices consist of similar device properties, the underlying algorithms can vary and thus produce different results. Mogavero et al. (2025) and Y. J. Lee et al. (2025) highlight that, as manufacturers use different training data, different selection processes and scoring methods, researchers should be cautious of generalising results without information on the proprietary algorithms. Consequently, limited disclosure of the algorithms makes evaluating the underlying reasons for varying results unclear.

A third methodological issue concerns the population variability. The populations consumer wearables are evaluated on influence the reported performance of devices. De Zambotti et al. (2024) argue that the current performance evaluation landscape in research is biased, with most studies performed on healthy individuals rather than including individuals with sleep disorders or minorities. As highlighted by Mahir et al. (2025), chronotype, gender, age and seasonal changes affect sleep patterns. Because sleep is highly individual, findings derived from limited study populations may not be generalisable to the wider population. This challenge is illustrated by Baron et al. (2018), who highlight that consumer wearables perform worse on populations with sleep disorders. As demonstrated by Mahir et al. (2025), patients with mood disorders showcase larger variation in sleep onset, offset and TST than healthy control groups. This highlights that measurement reliability depends on the characteristics of the individual studied. Moreover, populations with sleep disorders are prone to high error rates in sleep measurement with consumer wearables (Rentz et al., 2021). This is because algorithms are developed to identify physiological signals from healthy individuals, leading to high error rates in wearable outputs when physiology changes. As devices are predominantly evaluated on healthy individuals, the reported performance may overstate accuracy for the broader population.

The challenge does not only stem from the individuals studied in evaluation studies, but also from how well devices are able to capture physiological signals from different individuals. Ryals et al. (2023) note that physiological factors such as skin tone and thickness, hair, tattoos, sweat, and motion affect PPG measurements. They highlight that a combination of factors, such as obesity and skin tone, can have a larger effect on the signal. The effect of individual physiological factors on a device's ability to accurately measure sleep was demonstrated by Ajmal et al. (2021), who found that PPG signal quality was reduced by 61.2% in individuals with obesity and dark skin tone. These findings underline that the reported performance is influenced not only by the device but also by the population studied, as devices do not perform equally well on all users. This further suggests that the reported performance may not generalise to individuals who differ physically from the tested population.

The environment in which sleep data is collected is a fourth methodological factor that affects the reported performance of wearable devices. Consumer wearable data are collected in naturalistic environments, whereas reference measures are obtained clinically in controlled settings. De Zambotti et al. (2024) argue that this mismatch introduces a structural limitation, as laboratory performance does not necessarily translate to real-life conditions. This implies that lower agreement between wearables and clinical references does not directly indicate inaccuracy in wearables but can be partly influenced by the measurement context. In addition, naturalistic environments introduce external factors that contribute to errors in sleep measurement that are not present in laboratory settings (Rentz et al., 2021). For instance, the movement of bed partners can be falsely registered as the user's movement, thereby affecting measurement accuracy.

Beyond these factors, the evaluation method itself influences the reported accuracy of consumer wearables. Y. J. Lee et al. (2025) relate the variation between PSG and consumer wearables to the comparison of two structurally different processes in generating results. In PSG, physiological signals are scored manually by trained sleep technicians, whereas in consumer wearables, signals are interpreted automatically by proprietary algorithms that can vary between manufacturers. Van Gorp et al. (2022) extend this point by arguing that human scorers can recognise when data is faulty and handle it accordingly, whereas algorithms that are not programmed to handle such cases will still produce an output in situations when the result should be inconclusive.

Subsequently, the inherent limitations of PSG itself, when used as a reference standard, should be acknowledged. PSG as a method is not free from uncertainty. Van Gorp et al. (2022) argue that some sleep stages are scored

more consistently than others and that the average inter-rater agreement is 82.6% between human scorers. For instance, sleep stage N1 has an inter-rater agreement rate of approximately 50% (Ghorbani et al., 2022), whereas REM can reach 90.5% (Van Gorp et al., 2022). This indicates that the sleep measurements produced by PSG also contain variation depending on the stage being scored and the scorer. Rentz et al. (2021) argue that this variation, arising from inter-technician inconsistency, is carried over into training models and datasets. Consequently, consumer wearables are benchmarked and trained against a reference that is not free of variability, which can affect the perceived accuracy of wearables if inaccuracy is passed on.

Finally, Younes et al. (2022) acknowledge that the existing sleep scoring system is outdated and that recent technological advancements face substantial limitations with current standards. They note that the scoring rules were developed a decade ago with minor modifications. This reinforces the idea that the evaluation benchmark and methodology themselves might be factors constraining the reported accuracy of consumer wearable measurements.

Improving accuracy

Given these performance limitations, recent research and device developments have pursued two main approaches to improve agreement with PSG. These include algorithmic improvements, such as AI-enhanced models and personalised scoring, and hardware improvements through the integration of multiple sensing modalities (Bianchi, 2018; Mathunjwa et al., 2025; Perez-Pozuelo et al., 2020).

Mathunjwa et al. (2025) recognise that a key driver of improved performance is developments in AI-enhanced sleep analysis. Deep learning models enable the integration of multimodal data from multiple sensors, allowing for more complex representations of sleep patterns. Specifically, using deep learning models on PPG and respiratory signals has demonstrated significant improvements in the prediction accuracy of sleep stages (Kazemi et al., 2024). These models have been found to improve sleep-stage classification by processing inputs more effectively than traditional models, leading to reduced false detections and increased accuracy (Mathunjwa et al., 2025).

Perez-Pozuelo et al. (2020) argue that AI-based systems using machine learning are enabling personalised evaluation of sleep data. Through the utilisation of machine learning, scoring and analysis can be adjusted to the individual's data to increase algorithm performance. This was demonstrated in practice by Grandner et al. (2023), who adapted an algorithm personalised

to an individual's sleep data, resulting in improved specificity (wake detection) of over 10% compared with a generalised algorithm. Improvement in wake detection is an especially valuable finding, as it is a metric that wearables struggle to detect accurately.

The second approach concerns hardware improvements, where combining multiple sensing modalities into devices has been shown to increase performance (Mogavero et al., 2025). A direct demonstration of this is the increase in four-stage detection accuracy from 57% to 79% when combining accelerometer, temperature, heart rate variability and circadian features instead of relying solely on accelerometry (movement-based sensing) for sleep staging (Rentz et al., 2021). Having multiple layers of information helps with classification, as some sensor signals may miss information that others capture (Mathunjwa et al., 2025; Rentz et al., 2021).

Despite the advancements, consumer devices remain imprecise in comparison with the gold standard, especially for sleep stage detection (Mathunjwa et al., 2025). Djanian et al. (2022) further argue that much of the potential in the approach of using deep learning and AI models for sleep evaluation remains unexplored. Thus, while wearables are showcasing increasing performance, further research on their performance is needed as the technology develops.

In contrast with a broader body of literature that emphasises the need for further validation of consumer wearables (Birrer et al., 2024; Kainec et al., 2024; T. Lee et al., 2023), Van Gorp et al. (2022) challenge the emphasis on further improving accuracy in wearable and automated sleep staging. The authors argue that there are inherent uncertainties in sleep staging that create a fundamental limit on the achievable accuracy of agreement in sleep staging. As even expert scorers do not reach perfect inter-rater agreement, it follows that automated models cannot either obtain perfect measurement to the same standards. Lujan et al. (2021) similarly report an upper bound in achievable accuracy due to wearables' inability to fully replicate PSG measurements. The authors argue that this is because the signals measured by wearables and PSG are fundamentally different, and thus, peripheral signals from wearables may never capture sleep accurately. Consequently, Van Gorp et al. (2022) propose that emphasis should not be placed on large efforts to maximise accuracy, but rather on shifting focus to measuring the uncertainty in device predictions.

4.2 Transparency and interpretability

This section examines the underlying systems in consumer wearable sleep technologies. It focuses on the black-box nature, the limited visibility of the algorithms and the resulting challenges for interpretability of device outputs. In doing so, it reviews how the system's opacity affects understanding of the technology used and the metrics produced.

Black box systems

The literature widely considers CWT a black box (Baumert et al., 2022; Tuovinen & Smeaton, 2019). This stems from the system's internal logic being hidden from the user (Guidotti et al., 2018). In CWTs, this results in only the input and output being observable, while the algorithm and the system's computation remain unknown. Guidotti et al. (2018) recognise that black-box systems result in research on the topic examining either the input or the output. Furthermore, they highlight that implementing black-box systems is a risk because not knowing how the decisions are generated can have implications for ethics, safety, accountability, and liability.

Guidotti et al. (2018) argue that there is a need to implement principles that open up how black-box systems make decisions. However, by doing so, they recognise that this comes at the cost of interpretability and accuracy. They mean that, due to the complex nature of the systems, opening the internal logic would simplify it to an interpretable level, which would consequently influence the accuracy of the output (Guidotti et al., 2018). Tuovinen & Smeaton (2019) similarly recognise a convenience and transparency trade-off in wearables, as devices deny access to the logic of analysis or input data, creating a convenient product for the user but also resulting in an opaque system that raises ethical concerns.

Proprietary algorithms and limited transparency

Consumer wearables rely on proprietary algorithms to produce sleep measurements (Cho et al., 2021). As a result, they function as opaque systems in which key aspects of the data-processing pipeline, including sensor capabilities, data extraction, and computational methods are not publicly disclosed (Birrer et al., 2024; de Zambotti et al., 2024; Rentz et al., 2021). Consequently, there is a lack of understanding of how physiological signals translate into sleep metrics, affecting the validity and interpretability of the results (Rentz et al., 2021).

The proprietary nature of the systems extends into multiple layers in how consumer wearable sleep data is processed (Rentz et al., 2021). Rentz et al. (2021) mention steps, such as the sensing capability of sensors, data extraction, data cleaning, and processing, as stages that are invisible to users.

These processing steps are at the core of what consumer wearable outputs represent. The limited visibility and proprietary nature of CWT are widely considered core challenges in their validation and reliability (Baron et al., 2018; Birrer et al., 2024; Mathunjwa et al., 2025). Birrer et al. (2024) illustrate the scale of the issue by identifying widespread methodological challenges related to proprietary algorithms, with only 17 % of PPG-based devices disclosing features of the algorithms. This lack of disclosure limits independent validation of algorithm performance and accuracy (Mathunjwa et al., 2025), as well as limits clinical applicability and comparability across devices.

The transparency issue extends to limited accessibility to raw data. De Zambotti et al. (2022) and Roos & Slavich (2023) highlight that being unable to access raw data is a concern for both the reproducibility of studies and the interpretation of algorithms. As a result, Roos & Slavich (2023) highlight that researchers are constrained to use computations derived from proprietary algorithms (Birrer et al., 2024). Mahir et al. (2025) similarly note that a majority of studies reported findings based on proprietary algorithms. Reliance on manufacturer-produced data is further illustrated by Mahir et al. (2025), who report using data derived from proprietary algorithms and the resulting sleep and wake labels derived by the manufacturer. This demonstrates that independent validation is difficult, as there is a great reliance on data that manufacturers choose to provide. Consequently, the extent to which the findings reflect reality becomes uncertain.

Roos & Slavich (2023) critically note that wearable device-derived physiological data are often perceived as objective, even though they are shaped by subjective decisions in algorithmic development, data collection, and processing, which can influence the interpretation. This notion is especially relevant to metrics produced by manufacturers that lack a clinical equivalent, such as readiness, restlessness, and sleep scores. The accuracy and utility of such CWD-derived scores cannot be clinically validated, as there is no clear definition of what the scores incorporate and lack correlation to any patient-reported outcomes (Baumert et al., 2022).

The lack of accessible raw data also makes it challenging to benchmark devices (Imtiaz, 2021). In response to these challenges, authors have proposed establishing open-access databases and benchmarking datasets to increase data availability and comparability (de Zambotti et al., 2022; Mathunjwa et al., 2025). Such databases aim to address the challenge of comparability across devices and over time, which are topics examined in more detail in the following sections.

4.3 Technological evolution and long-term reliability

The following section examines the comparability and long-term continuity of sleep metrics and measurements from CWDs. Given that these technologies are often used to measure sleep over long periods in naturalistic environments, their value depends on the assumption that measurements remain stable and comparable over time, across studies and devices. The section first reviews how wearable technologies have evolved. Then, considers the long-term measurement of sleep with wearables, and finally, discusses the challenges of comparing wearable devices and their measurements.

Technological evolution

Consumer wearable technology has developed rapidly in both hardware and software in recent years. Research suggests that newer devices incorporating multiple sensing modalities can provide more accurate and comprehensive sleep measurements (Mogavero et al., 2025). The literature on developments in consumer wearables is mainly framed by performance improvements reported in evaluation studies. However, less attention has been paid to the technological change itself, such as continuous updates, algorithm modifications and their implications for device outputs and interpretability of results.

Mogavero et al. (2025) position consumer wearable sleep as a young field, with devices first emerging in the early 2000s as fitness trackers relying on a few or a single sensing modality, such as accelerometry. In the 2010s, devices began integrating multi-sensor systems, enabling more complex data collection, which was followed by the development of more complex computational approaches. Lujan et al. (2021) and Mogavero et al. (2025) indicate that from an algorithmic perspective, the evolution represents a shift from movement-based accelerometry to the integration of deep learning models. The authors highlight that the use of deep learning models has enhanced processing capabilities and enables the detection of small fluctuations in physiological signals by learning how one state in a sleep epoch influences the next epochs.

The evolution of wearable sleep technology has been rapid. Guillodo et al. (2020) note the rapid evolution of technology as a reason research struggles to keep pace with commercial releases. Similarly, Kainec et al. (2024) note that software updates are introduced nearly every month. This is exemplified by the Oura Ring 4, which has already had 20 updates since its initial release in October 2024 (Oura, n.d.). The rate at which new releases are introduced is illustrated by Doherty et al. (2024), who note that studies become

historical as soon as they are published. Furthermore, they note that all the wearables reviewed in their study had been replaced by newer models or had been discontinued by the time the paper was published.

Birrer et al. (2024) recognise that underlying changes in algorithms are not as widely addressed in the literature as the number of studies examining wearable device use. Accordingly, the authors articulate that algorithm development and its associated challenges remain largely unexplored. Baron et al. (2018) and Imtiaz (2021) support this perspective by highlighting that most research on consumer wearables focuses on performance evaluation of devices. When considered together with the limited number of long-term studies noted by Guillodo et al. (2020), this suggests that the existing literature may provide a limited basis for examining how consumer wearable algorithms have evolved longitudinally.

Moreover, Mogavero et al. (2025) highlight that the evolution of integrating multi-sensor capabilities into wearables has greatly increased the ability to understand sleep, as it allows for continuous long-term data collection. However, they also notice that increased complexity introduces new challenges, such as combining signals with varying sampling frequencies, noise levels and motion sensitivity. These factors introduce computational complexity, while the barrier of transparency still limits understanding of the decision-making process. Likewise, Ghorbani et al. (2022) and Óskarsdóttir et al. (2022) welcome advancements but press that improvements raise concerns about long-term data harmonisation and reliability, as many challenges remain unresolved. While algorithmic changes represent technical improvements, they create fundamental challenges to continuity and the reliability of long-term sleep tracking through wearables, which is further explored in the following section.

Longitudinal measurement

Consumer wearables are intended to be used continuously in everyday environments. The devices are set apart from clinical measurement tools by allowing large-scale real-world data collection, which provides a foundation for understanding sleep behaviour with broader health outcomes (Mogavero et al., 2025). To understand the implications of sleep on health and diseases, longitudinal, accurate, reliable and efficient sleep measurement is needed (de Zambotti et al., 2024; Pan et al., 2020). Longitudinal research on the use of wearables in naturalistic settings is thus significant for providing insights into their reliability in the intended use cases. This section examines how literature currently examines long-term measurement with wearables.

Even though consumer wearables have the potential to provide great value for long-term monitoring in naturalistic environments, there is a limited number of studies examining this. Baron et al. (2018) highlight that there is a lack of research into whether consumer wearables are accurate enough to provide insights on a population level and recognise sleep pattern changes over time. Moreover, Guillodo et al. (2020) recognise the absence of long-term studies on consumer wearables, noting that sleep measurement using wearables in long-term settings has been far less explored than in short-term settings. Both Baron et al. (2018) and Guillodo et al. (2020) urge for more longitudinal studies, as longitudinal sleep recording can reveal relations between sleep quality and other health outcomes. Taken together, these findings suggest that the understanding of wearable use in long-term settings is significantly limited by the absence of such longitudinal studies.

Two longitudinal studies provide more direct evidence of the required considerations for long-term measurement. Óskarsdóttir et al. (2022) express that it remains uncertain whether longer measurement periods yield a more thorough sleep-stage picture compared to short monitoring periods. The authors addressed this gap by analysing sleep, physical activity and heart rate over six months using actigraphy (movement-based sensing). They found that a 3–14-night period, commonly used for sleep measurement, is insufficient to capture reliable sleep habits. Their findings suggest that measurement windows exceeding two months provide more sufficient results, due to the high variability observed in individual sleep patterns. Leota et al. (2026) elaborate on this by differentiating between the dimension of variability being measured. When addressing average variability in sleep, 2-7 nights were sufficient, whereas when addressing sleep variability in consecutive nights, a significantly longer measurement window was needed, spanning 45-65 nights. The studies agree on the conclusion that longer measurement periods than the current standard practice are necessary when using wearables to reliably address variability and characteristics in sleep, and that the required length depends on what part of sleep is being examined (Leota et al., 2026; Óskarsdóttir et al., 2022). Consequently, short monitoring windows that currently dominate sleep research might provide unreliable information on sleep variability, even though devices would perform with high accuracy.

An additional perspective concerns how longitudinal sleep data should be analysed. Óskarsdóttir et al. (2022) suggest that analysis of measures should be conducted through an individualistic approach to understand sleep patterns over time. This is motivated by high variability in sleep from person to person, and noting that an individualised approach can yield more valuable results over time, in contrast to combining data from multiple people to address group-level trends. An individualistic approach may also

be relevant from a measurement perspective, as individual physiological factors affect the device's signal quality and consequently the results (Ryals et al., 2023). A great shortcoming of sleep literature is that most literature evaluates single nights' sleep or a combination of nights from several individuals, rather than measuring longer periods where sleep patterns can be identified for individuals (Óskarsdóttir et al., 2022).

Collectively, longitudinal measurement with wearables has not been sufficiently covered in the literature, despite its central role in identifying sleep patterns and potentially linking them to health outcomes. The existing evidence points toward measurement windows significantly longer than current practice and toward an individualistic focus that enables recognition of individual patterns rather than cross-comparison. As longitudinal studies emerge, it is necessary to ensure that measurements obtained over extended periods remain stable. If measurements cannot be perceived as equivalent across time, the value of interpreting longitudinal measurements is diminished.

Comparability dimensions

Comparability is closely tied to long-term measurement reliability. If sleep metrics produced by consumer wearables are not stable or reproducible over time, the ability to draw meaningful comparisons from longitudinal data is undermined. The stability of sleep measurements can be considered through three dimensions: whether measurements from different devices are comparable, whether measurements remain stable within the same device across versions, and whether measurements remain stable in a single device across repeated use over time. This three-dimensional structure is illustrated in Figure 2, which showcases the relationship between the levels of comparability. The dimensions should be interpreted as a nested structure rather than three separate problems. Instability in within-device metrics affects the reliability of comparisons across device versions, which in turn undermines comparisons across different devices.

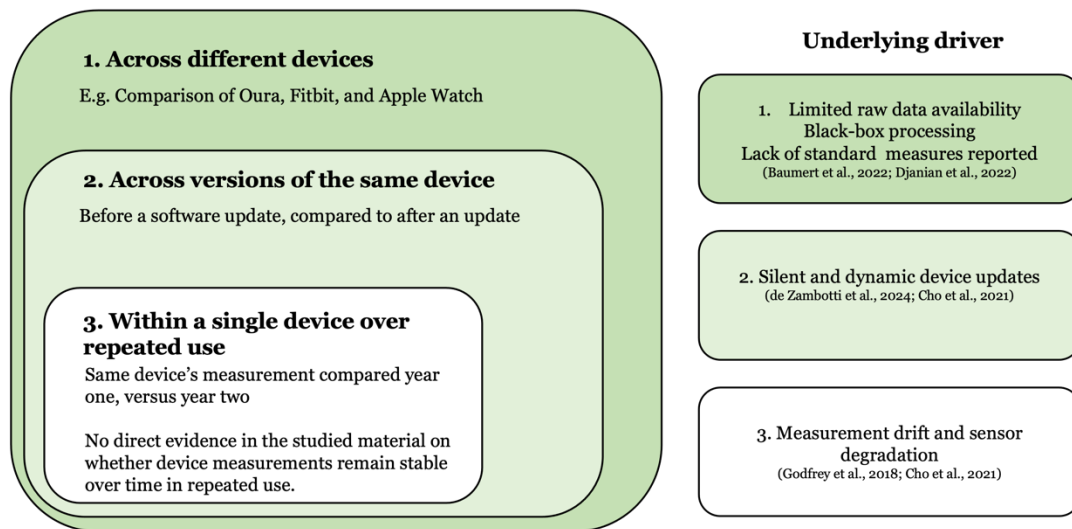


Figure 2: Three levels of comparability in consumer wearable sleep metrics, consisting of across devices, across device versions, and within a single device. Levels are related to the underlying drivers affecting comparability.

In each case, the underlying causes trace back to challenges highlighted in section 4.2, proprietary algorithms limited raw data availability, compounded by dynamic device updates (de Zambotti et al., 2024). Comparability of measurements is an essential factor for consumer wearables because devices are intended for long-term use, meaning changes in measurement outcomes should reflect changes in the person being measured rather than in the measurement instrument.

The literature examined does not explicitly frame comparability of wearable metrics as a distinct research area comparable to performance evaluation studies. Instead, the topic arises indirectly through evaluation and the limited reproducibility caused by the opaque nature of wearables (Roos & Slavich, 2023). Additionally, there is a lack of studies examining the longitudinal comparability of sleep metrics.

The most directly addressed level concerns measurements across different devices. Baumert et al. (2022) argue that objective comparison between sleep patterns and sleep architecture from different devices is currently impossible because algorithms are not described in detail and data is processed by black-box systems. Djanian et al. (2022) further highlight that comparison is undermined by a lack of agreement on what measurements to report, and additionally, that the number of sleep stages a device is evaluated on affects comparison across studies. In practice, this entails that a device evaluated on only two stages of sleep cannot directly be compared to one evaluated on all five sleep stages. Interestingly, Baron et al. (2018) conclude that comparisons

between different wrist-worn wearable devices commonly yield similar agreement in sleep measurement, whereas comparisons with PSG yield moderate to poor agreement, with the authors suggesting that devices may share similar biases in sleep estimation.

In a systematic review of data quality, Cho et al. (2021) found that heterogeneity in wearable device data can be understood at three levels: within a single person using a single device, across multiple persons using the same brand, model, and version, and across multiple persons using different devices. The authors highlight that proprietary algorithms increase heterogeneity in multi-device data, making it difficult to compare results across devices, and that variability in single-person data also influences comparisons when measuring different people. Furthermore, Cho et al. (2021) illustrate that even though studies describe data quality challenges, they seldom expand on how quality metrics are defined and how the challenges were assessed. Thus, contributing to a wider methodological comparability challenge.

The second level concerns comparability across versions of the same device. This implies comparability of measurements produced before and after a change in the measurement instrument, such as a firmware, software, or hardware update, and whether these can be treated as equivalent. Ghorbani et al. (2022) conducted an evaluation study, where they demonstrated that applying a Gen3 algorithm retrospectively to Gen2 data in an Oura ring resulted in improvement in four-stage sleep stage classification with 14.1% mean accuracy compared with the older Gen2 algorithm applied to the same data. The study demonstrates how large the differences in results can be between new and old generations when applied to the same data. Consequently, data from the same device but processed by different algorithms might not yield directly comparable measurements.

Godfrey et al. (2018) importantly note that technological advancement is not linear, and that updates in devices can impose qualitative changes in circumstances that make drawing insights from past technologies difficult. For consumer wearables, this should be understood as improvements and potential discontinuities that can break compatibility across device updates. De Zambotti et al. (2024) support this perspective by arguing that frequent updates leave inconsistencies in the data. They advise that longitudinal studies should ensure stability of measurements by disabling updates, and that even in short-term studies, the hardware, firmware, and software should be reported, as results can become irreproducible after updates. An illustrative example is T. Lee et al. (2023), who froze software updates across all examined wearables in one of the largest simultaneous multidevice validation studies to limit data heterogeneity caused by updates.

Adding to the complexity of device updates affecting compatibility, Cho et al. (2021) note that updates are often silent, meaning that users are not always made aware of new software releases or when they are installed. In retrospect, it becomes difficult to assess whether changes stem from an update or from changes in the user's sleep habits. This may even affect the comparability of different individuals' sleep data if sleep has been measured on different time periods.

The third level concerns comparability within the same device over an extended period. That is, whether sleep measurements from a single device over repeated sessions are stable. This area has the least direct evidence in the reviewed literature, and no study examined in this review directly assesses whether wearable devices produce stable measurements over time under consistent conditions. De Zambotti et al. (2024) explicitly state that device precision over multiple measurements is poorly represented. Additionally, Mahir et al. (2025) suggest the consistency of measurements be further researched in the future. The closest evidence comes from Óskarsdóttir et al. (2022), who demonstrate that longitudinal sleep measurement is needed to access reliable estimates due to high individual sleep variability. Because the study did not explicitly separate between measurement variability stemming from devices or individual sleep variability, further research is needed to determine whether observed fluctuations stem from one or potentially both variables. Óskarsdóttir et al. (2022) propose an approach where data from the same device is analysed, as they see that the value of wearables lies in being able to capture longitudinal patterns. This proposed approach could potentially function as an approach to determine within-device measurement instability.

Godfrey et al. (2018), in turn, argue that wearable technology is prone to drift, referring to the accumulation of measurement errors from physical changes over time. The authors state that devices should have strategies for recalibration to reduce drift in measurement. Cho et al. (2021) similarly recognise changing measurement quality over time, but relate it to sensor degradation over time. Together, these observations suggest that the within-device stability of sleep measures over time is a relevant but unexplored topic in the literature.

Taken together, the three dimensions of comparability in wearables and sleep measurements reveal that comparability is not a well-defined problem in literature, but rather, comparability issues arise whenever results or device performance are placed in relation to each other. These challenges appear across different devices, across time and across device versions. Baumert et al. (2022) highlight that longitudinal multiple-night measurement is one of

the key advantages of wearable devices, as it can reduce misclassification through longer measurement periods and enable better monitoring of health outcomes at scale. The longitudinal use of wearables is affected by the silent algorithm updates, measurement drift, and cross-device heterogeneity. In longitudinal use, these challenges become more prominent for the interpretability of results. At the core of the comparability challenges are proprietary algorithms and limited raw data access, which make it challenging to differentiate whether changes in observations stem from the device or the user (Cho et al., 2021). The complexity of comparison across measurements is further enhanced by the lack of standardised reporting standards.

A call for standardisation

With the growing use of consumer wearables in personal health management and research, numerous authors have expressed concerns for the lack of established reporting practices and common tools for evaluating, interpreting, and using wearables in research (Baron et al., 2018; Baumert et al., 2022; Birrer et al., 2024; de Zambotti et al., 2024; Doherty et al., 2024). Older reviews tend to express the problem directly, calling for common standards, whereas recent work incorporates the standardisation challenges with the broader environment influenced by rapid technological advancements, algorithmic updates, and more complex device systems.

Baron et al. (2018) highlight a lack of standardised criteria and agreement on what is acceptable performance against PSG. They recommend implementing a standardised validation system of consumer wearable technologies in relation to the setting studied to increase the comparability of studies and technologies. Their review indicates heterogeneity in validation approaches, with studies using a variety of different methods to evaluate wearable device performance. In this sense, Baron et al. (2018) frame the standardisation from an evaluation perspective concerning the interpretation of sleep measures.

Baumert et al. (2022) similarly connect standardisation to the comparison and interpretation of sleep patterns and architecture provided by wearable devices. They note that objective comparison of sleep patterns across devices is impossible to do and seek the establishment of standardised reporting practices, validation frameworks, and verification for products meeting the standards. Thus, Baumert et al. (2022) expand the standardisation topic from a comparison standpoint and argue for devices to adhere to standard metrics, minimum performance evaluation requirements against PSG, reporting practices, and benchmarking opportunities. They additionally

mention algorithmic fluidity as a challenge, but do not address it through standardised evaluation and resulting reporting protocols.

More recent literature builds on this concern but additionally expands it. De Zambotti et al. (2024) highlight that evaluation standards have been introduced, but there is a need for an operational framework that defines when devices are sufficiently validated and approved by experts. Furthermore, they express the need for studies to adhere to existing recommendations. They consider the standardisation challenge in the context of the evolving technological landscape that is affecting both device capabilities and modalities.

A similar shift is visible in Doherty et al. (2024), who emphasise the need for standardised validation protocols, and additionally that these should be thorough, adaptable, and take into account the evolving technological landscape. Similarly, to de Zambotti et al. (2024), Doherty et al. (2024) show that studies are failing to adhere to accepted gold-standard criteria for device evaluation and best practices for statistical analysis when determining device accuracy. Furthermore, the authors emphasise that heterogeneity in methodologies arising from not adhering to established frameworks is undermining the integrity of device assessment and limiting the applicability of wearables' results.

Birrer et al. (2024) build on this discussion by highlighting that there are no protocols for evaluating wearables that do not contain EEG sensors. They suggest more consistent reporting of used sensors and validation details to ensure transparency and comparison as the devices and field of sleep evaluation continue to evolve. Additionally, they see open-source classifier code as essential for cross-comparison. The authors highlight frameworks as especially important to be able to evaluate the benefits of more complex devices that include multiple sensing modalities.

Taken together, these articles suggest that standards for performance evaluation, reporting and comparison of consumer wearables are emerging and to some extent already exist. The authors agree on the need for widely adopted protocols to ensure valid performance metrics and allow comparison across studies and devices. The more recent literature additionally acknowledges that the standards need to consider consumer wearables in the context of the evolving technological landscape in which they exist (Birrer et al., 2024; Doherty et al., 2024). Thus, an emphasis should be on establishing adaptable standards that can react to the evolving technologies.

4.4 End-user implications and interpretation

Previous sections have established how wearable devices perform from a research perspective; this section shifts its focus to the end user. It examines the relationship between the metrics presented to users and what they represent. In addition, it considers how wearable sleep tracking affects users and how users should interpret wearable data. Addressing the challenges associated with wearables from a user perspective is essential, as they extend beyond individual users and affect millions of people globally (Baumert et al., 2022; Dion et al., 2025). This section first discusses the accuracy of wearables from an end-user perspective, then examines the longitudinal interpretability of sleep metrics, and finally outlines the behavioural implications of wearable use.

Accuracy and interpretation

As established in section 4.1, the accuracy of wearable sleep depends on the metric and the device examined. Overall accuracy for two-stage sleep and wake detection reaches approximately 87% agreement with PSG (Birrer et al., 2024). However, agreement drops when more stages are considered. Devices also exhibit overestimation of TST by up to 10% and poor detection of wakefulness (Doherty et al., 2024), and perform less reliably in non-healthy populations (Rentz et al., 2021). A tension arises from the situation that the devices have been widely adopted before achieving sufficient accuracy (Dion et al., 2025). This variability and uncertainty place the end user at a disadvantage when interpreting their own sleep data, because the factors influencing measurement accuracy are not visible in the metrics presented to them.

Transparency is important for users to assess the validity of data. Y. J. Lee et al. (2025) argue that, due to limited access to raw data and opaque algorithms, consumers should interpret metrics generated by wearables with caution and acknowledge their accuracy limitations. Kuosmanen et al. (2022) build on this idea, arguing that users struggle to assess measurement accuracy without transparency. Without knowledge of what results are based on, the validity is difficult to assess.

Tuovinen & Smeaton (2019) define the convenience of consumer wearables as a core trade-off users face when using them for sleep monitoring, due to the lack of transparency. The authors contrast the convenience provided by the devices with the ethical implications of being unaware of how the analysis is conducted, being denied access to one's own raw data, and not being made aware of the uncertainty surrounding the aggregated information presented. A concern is that results are presented as objective, even though they represent the device's approximation. This creates a grey area in determining

when users should be concerned with their own data, as the output is difficult to interpret if it does not clearly contradict the user's own perception (Tuovinen & Smeaton, 2019). Kuosmanen et al. (2022) add that users should receive greater transparency into why errors happen. However, increasing transparency might not yield better interpretation outcomes, as Tuovinen & Smeaton (2019) argue that users would find it more difficult to interpret the results, because they would not be presented in as simple a way. Additionally, users might not be sufficiently equipped to analyse the full data if manufacturers were to provide it.

At the core of user interpretation of wearable accuracy is a mismatch in reported and perceived accuracy. Baron et al. (2018) argue that in several cases, the claims about the device's performance outweigh the evidence supporting them. This means that users may perceive the results as more accurate and reliable than they are. Rentz et al. (2021) build on this idea, arguing that companies do not consider how users interpret device output. This creates a risk that users interpret data without reference to verified accuracy. As a result, if a result does not align with the user's perception, it can create doubts about other collected data and lead the user to distrust the system as a whole (Kuosmanen et al., 2022). Bianchi (2018) adds that sleep trackers do not provide users with an immediate way to validate data. This poses a risk that a user draws conclusions from inaccurate data, without a measurement error ever becoming visible.

The mismatch between accuracy and perception continues into the metrics users commonly pay attention to. Kuosmanen et al. (2022) note that sleep stages are most commonly a metric of interest when assessing one's own sleep quality. Schwartz et al. (2024) report a similar pattern in which TST and sleep staging are the most valued metrics. However, as Kuosmanen et al. (2022) argue, there is no direct evidence that behavioural changes affect the amount of time spent in different sleep stages or their timing. Given that sleep stage detection shows poor agreement with PSG, users may value metrics with variable accuracy. As Schwartz et al. (2024) also note, there is a pattern in which users are drawn to measures based on perceived importance rather than clinical relevance. This can also be seen in users' preference for scientifically evaluated devices endorsed by academic, medical, or government institutions. It can be concluded that users are paying attention to metrics on the assumption that they are clinically valuable, but whose accuracy may be low compared to clinical standards. This raises a significant concern about the use of wearables for sleep measurement, as the data users draw insights from may lack verified accuracy.

Taken together, wearable results contain variation that must be acknowledged and accepted when forming judgements. This uncertainty

relates to both the underlying system and users' difficulty in interpreting the presented aggregated results. In longitudinal use, these uncertainties become more important because users compare results over time.

Longitudinal tracking of personal baselines

Wearables are particularly promising for naturalistic, large-scale longitudinal data collection (Perez-Pozuelo et al., 2020). Baumert et al. (2022) highlight that users' longitudinal drawing of conclusions from their sleep data has profound global implications due to the scale of wearable use. Because the data vary widely from night to night, the value lies in observing sleep patterns over time rather than interpreting individual nights.

As highlighted by Óskarsdóttir et al. (2022) and Leota et al. (2026), longer measurement periods can yield a better understanding of sleep when measured with wearables. In addition to sleep naturally varying from night to night, users should aim to interpret patterns rather than focus on specific metrics or individual nights' results. In practice, this means that the reference for users is personal historical sleep data. Kuosmanen et al. (2022) highlight that users commonly compare results daily with previously obtained results. The authors highlight that this monitoring leads to competing with the user's own historical data. However, unlike researchers with access to PSG, users are less equipped to assess the limitations of the results.

While Clawson et al. (2015) note that mistrust in devices challenges long-term adoption, Nelson et al. (2020) found that users continued to wear devices despite doubts about their accuracy. Nelson et al. (2020) describe device feedback as either confirming or invalidating and note that, over time, individuals become increasingly aware of perceived discrepancies. The perceived mistrust stems from mismatches between perceived and reported data. As sleep scores and sleep can exhibit substantial variability (Óskarsdóttir et al., 2022), long-term measurement is needed to reliably capture patterns. However, such long-term use can undermine sustained adoption and cause mistrust of the device, creating tension in the longitudinal use of wearables.

These findings suggest that long-term use could be supported by increasing device transparency, enabling users to gain a clearer understanding of results and errors. Since sleep varies over time, long-term use can be valuable, but users need to trust and understand the data provided. Transparency about device limitations could help users assess the reliability of results and maintain trust. However, greater transparency should not lead to more complicated data to interpret.

Behavioural implications

Research has largely focused on clinical evaluation rather than real-world user implications (Dion et al., 2025). The literature indicates that the use of wearables might not only provide positive insights but also involves concerns about overall well-being, with self-monitoring sometimes being counterproductive (Baumert et al., 2022). Understanding these behavioural implications is therefore essential to evaluating the role of wearables in sleep measurement for end users.

Even though the overall accuracy of devices remains unverified, users report a positive effect on sleep by using wearables (Dion et al., 2025). Kuosmanen et al. (2022) found that most individuals who used wearables felt that wearables helped maintain a stable sleep rhythm and improve sleep quality. Wearables also support the improvement in sleep routines and can increase self-awareness of one's sleep habits (Klier & Wagner, 2022; Mathunjwa et al., 2025). Klier & Wagner (2022) argue that when users interact with device feedback, their sensitivity to their own sleep behaviour increases, enabling more effective lifestyle changes and earlier detection of unhealthy sleep disturbances. In this sense, wearables provide an affordable and usable way for general users to track their sleep (Mathunjwa et al., 2025).

A concern about wearable use is the negative effects. Mathunjwa et al. (2025) express concerns about over-reliance on technology, sleep anxiety and excessive self-monitoring. Dong (2025) similarly found that users experience anxiety when their smartwatch data is abnormal, with 40.7% of users reporting anxiety for abnormal data, and 45.6% experiencing dependence on sleep status. Even though a large majority of users are neutral about sleep results, Kuosmanen et al. (2022) highlight that device use also gives rise to users who aim to perform well in numerical metrics. This can cause sleep-related anxiety driven by an aim to perfect and 'perform' sleep. Excessive fixation on sleep data is termed orthosomnia (Baron et al., 2017).

The perceived effects of wearables should be considered in terms of the types of users who use the devices. Users of wearables exhibited longer sleep latency (the time it takes to fall asleep), one hour less TST, and worse insomnia symptoms than non-users (Dion et al., 2025). An underlying reason might be that individuals with sleep disorders tend to use wearables for sleep monitoring more often. As wearable performance in non-healthy individuals is worse and remains poorly researched, these findings should be interpreted with caution.

Taken together, wearables exhibit both positive and negative implications for end users. While the devices' aim is to support well-being, overreliance on the technology and anxiety about abnormal data can be counterproductive. This points to a significant need for devices to have both validated accuracy and increased transparency, so that users can interpret the data and uncertainty about what the metrics represent can be mitigated. Potentially, this could lead to the negative behavioural implications being minimised.

4.5 Summary

Table 1 summarises the prevailing knowledge in the literature across the four sections in this chapter. In addition, it identifies the gaps and challenges within the examined literature. The table is structured around the four key themes that address each research question.

Table 1:

Identified knowledge and challenges in the academic literature and discussed themes.

Research area	Prevailing knowledge	Identified challenge	Key sources
Performance evaluation and accuracy (RQ1)	Performance is metric and device specific with significant variability in measurement accuracies. Methodological aspects, such as device properties, transparency, population variability and the reference standard affect the reported accuracy. Accuracy is improving through AI-enhanced models and multi-sensor integration.	Evaluation studies predominantly utilise PSG as a reference standard, even though measurements and approaches are not equivalent. Methodological limitations are acknowledged, but non-healthy populations and studies in naturalistic settings remain underrepresented in literature.	(Birrer et al., 2024; Y. J. Lee et al., 2025; Lujan et al., 2021; Mathunjwa et al., 2025; Mogavero et al., 2025)
Transparency and algorithmic development (RQ2)	The underlying system functions as a black-box and is a core challenge for interpretability, independent validation, and comparability of results. Technological developments are rapid but risk creating more complex systems that further undermine interpretability.	Algorithm changes and their effects on results are rarely addressed in the literature. There is a need for widely adopted reporting and methodological standards, as well as consistently adhering to these standards.	(Birrer et al., 2024; Ghorbani et al., 2022; Guidotti et al., 2018; Mogavero et al., 2025; Rentz et al., 2021; Roos & Slavich, 2023)
Continuity of measurement results (RQ2)	Longitudinal use is a key use case for wearables, due to high variability in individual nights. However, the continuity of measurements cannot be verified, as the contents of device and algorithm updates are often undisclosed.	Comparability is discussed indirectly when results are placed in relation to each other. Literature does not examine comparability on its own and what the comparability challenges are for usability of wearables in sleep monitoring. The limited number of longitudinal studies significantly limits the ability to assess comparability across devices, updates and over time.	(Baumert et al., 2022; Cho et al., 2021; Ghorbani et al., 2022; Godfrey et al., 2018; Óskarsdóttir et al., 2022)
End user implications (RQ3)	Wearables support sleep awareness and habits but can cause over-reliance and dependence, resulting in anxiety. With limited access to a verified reference, the perceived accuracy of results has significant weight on users' attitude toward the device.	Studies discuss the mismatch between perceived and reported accuracy but provide limited guidance on how users should relate to presented information.	(Baron et al., 2017; Bianchi, 2018; Dion et al., 2025; Kuosmanen et al., 2022; Mathunjwa et al., 2025; Rentz et al., 2021; Tuovinen & Smeaton, 2019)

5 Discussion

This chapter concludes the thesis by synthesising the key findings from the literature on the accuracy and longitudinal use of wearable devices for sleep measurement. The study was guided by three research questions: How are sleep metrics from consumer wearables evaluated, and how accurately do they measure sleep? To what extent are changes in algorithms and hardware in wearable devices considered in the validation and long-term comparability of sleep metrics? And based on these factors, what are the implications for end users using the devices? To assess these questions, this section first outlines the main findings for each question (5.1). This is followed by a discussion of the thesis's reliability and limitations (5.2) and concludes with directions for future research (5.3).

5.1 Findings

This thesis examined the validity and long-term stability of sleep metrics produced by consumer wearables by reviewing the literature on performance evaluation, transparency, technological change, and end-user implications. The main finding of this thesis is that consumer wearables show potential for sleep monitoring in naturalistic settings and for longitudinal use, but their outputs remain variable, thereby limiting their clinical equivalence to PSG. The thesis suggests that uncertainty in wearable sleep metrics is twofold, stemming from both the device and methodological considerations in wearable research. Limited device transparency and a lack of standard practices in the research field complicate interpreting accuracy and assessing longitudinal implications. Subsequently, the end user is the one ultimately facing uncertainty and left to interpret the results in relation to their own perception of sleep, without clear insights into the underlying system or the results.

Regarding the evaluation and accuracy of wearable-derived sleep measures (RQ1), the literature indicates that wearable sleep metrics are predominantly evaluated by comparison with PSG. The literature suggests that devices are relatively good at detecting two-stage sleep and wake classification, reaching 80%-90% agreement with PSG, but are significantly less precise with more detailed sleep stage classification. The accuracy depends on the metric and device examined, rather than being constant across wearables. Devices consistently show systematic biases, overestimating TST and underestimating periods of wakefulness. These findings suggest that wearables can be used for approximate sleep patterns, but the accuracy remains insufficient for precise sleep measurement.

The review shows performance is shaped by several methodological and technical factors. These include sensor fidelity, algorithm transparency, population variability, measurement environment, and PSG as a reference standard. As most evaluation studies are performed on healthy individuals in controlled settings, the accuracy of wearable devices may not generalise to all users and real-world contexts. Furthermore, because PSG as a reference standard itself contains inter-rater variability, this uncertainty should be considered when wearable metrics are compared against the gold standard. Reported improvements in wearable accuracy, compared with PSG, must thus be considered in relation to the possibility that they are partly attributed to variability and changes in methodological and environmental factors, rather than to improved device accuracy.

Regarding the extent to which algorithmic changes and updates are considered in validation and long-term comparability (RQ2), the review finds that the proprietary nature of wearable technology poses a core challenge for interpreting and comparing device outputs. The reviewed literature indicates that changes in underlying technology are not sufficiently considered in the long-term comparability of results. Consumer wearables are described as black-box systems in which only the input and output are observable, and even the raw sleep data is often proprietary. This limits understanding of when changes in metrics reflect changes in the user rather than the system. Even though the literature acknowledges algorithmic changes and the need for consistent reporting practices, none of the reviewed articles provides a clear framework for evaluating the longitudinal stability of wearable metrics.

The findings suggest that the challenges of limited transparency become more prominent when measures are compared over time. Algorithm updates occur frequently and introduce uncertainty, as the contents of the updates and their effects on the comparability remain unknown. The thesis suggests that comparability can be understood on three levels: between devices, across device versions and within a device over time. Currently, the literature discusses comparability indirectly through algorithm opacity, limited data access, updates, and sensor degradation, but does not directly examine measurement stability over time. This is a central gap because longitudinal use is both an advantage of the devices and the way they are used.

Regarding the end-user implications (RQ3), the literature shows a tension between how sleep measures are presented to users and how users interpret them. Devices present metrics as objective, even though they represent the algorithm's best approximation, without being transparent about the level of uncertainty associated with the result. Users overvalue clinically framed

metrics, such as sleep stages, whose accuracy remains limited. The reviewed literature suggests wearables can support sleep routines and sleep quality, and increase awareness of one's own sleep. However, they can also lead to anxiety and dependence on devices, resulting in over-reliance on the technology, which can lead to a fixation on sleep scores. Because users lack information on what measures represent and how the system functions, they are poorly equipped to assess the reliability of sleep measures.

Taken together, the themes addressed by the three research questions should be considered as interdependent challenges rather than separate problems. The accuracy of wearable sleep metrics cannot be assessed without considering the opacity of the systems that produce them, and longitudinal stability cannot be assessed without considering technological changes, as changes can affect continuity of results. End users face this uncertainty as a gap between perceived and verified accuracy. The central challenge this thesis assesses is that, while wearables have great potential for measuring sleep in large-scale, longitudinal, naturalistic environments, in these settings, the challenges of limited transparency, device updates, unverified measurement stability, and difficulty interpreting results become most evident. Wearables should thus be evaluated not only on their accuracy relative to PSG, but also on whether their measurements are stable, comparable, and interpretable over time. Based on the findings in this thesis, while the accuracy of short-term use of wearable sleep measures is more established, the stability over long-term use remains insufficiently understood. The literature indicates that several factors need to be addressed to understand the continuity of results and comparability over time.

5.2 Limitations and reliability

The reliability of the thesis is fundamentally affected by the limited scope of the reviewed literature. This review examined 47 scientific papers. However, the thesis covers four broader themes: device evaluation and performance, transparency, comparability, and the user perspective on wearable use for sleep measurement. Because of this scope, the thesis is only able to provide a partial perspective on the broader phenomenon. This implies that individual articles carry significant weight in the conclusions, and including additional articles would yield a different perspective on the current research field.

While recent literature was prioritised, the rapid technological evolution means that the thesis only captures a snapshot of the current state of wearable devices available on the market. However, it still provides a direction for what researchers should consider and how users should relate to wearable devices for sleep measurement. Because the article selection was

focused on 2018 onward, some articles are less established and certain areas, such as longitudinal studies and how algorithm updates affect long-term reliability, have less available literature. Conclusions drawn from these areas should be treated with caution.

Another limitation is not systematically distinguishing between clinical-grade and consumer-grade wearable devices, specific models, and the sensing modalities used when assessing device accuracy, performance, longitudinal stability, and user perceptions. This affects the conclusions of measurement accuracy, as performance can vary widely depending on the device and its features. Finally, the thesis did not use third-party involvement or conflicts of interest as selection criteria when selecting articles. This may affect the interpretation of some studies, as manufacturers may publish articles that demonstrate high agreement with clinical standards to present the device in a positive way. This limitation underlines the need for independent and transparent evaluation studies.

5.3 Future research

This thesis has examined the longitudinal validity and stability of wearable sleep metrics. However, significant gaps remain. Continuous monitoring is the intended use case for wearables, yet the research field lacks long-term studies of accuracy and variability, and of the effects that black-box computing has on the interpretation of results and on end users. Future studies should examine how changes in the device's hardware and software affect the continuity and accuracy of results in longitudinal use. Furthermore, comparability across devices and studies is undermined by inconsistent methodologies and reporting practices. Addressing these areas is critical to understanding the use of wearables as sleep-measuring tools. Future research should examine interdependencies of the challenges, rather than view them as separate issues. By addressing these gaps, future research could contribute toward more reliable standards for evaluating and interpreting consumer wearable sleep metrics in longitudinal settings.

References

- Ajmal, Boonya-Ananta, T., Rodriguez, A. J., Du Le, V. N., & Ramella-Roman, J. C. (2021). Monte Carlo analysis of optical heart rate sensors in commercial wearables: the effect of skin tone and obesity on the photoplethysmography (PPG) signal. *Biomedical Optics Express*, 12(12), 7445–7457. <https://doi.org/10.1364/boe.439893>
- Baron, K. G., Abbott, S., Jao, N., Manalo, N., & Mullen, R. (2017). Orthosomnia: Are some patients taking the quantified self too far? *Journal of Clinical Sleep Medicine*, 13(2), 351–354. <https://doi.org/10.5664/jcsm.6472>
- Baron, K. G., Duffecy, J., Berendsen, M. A., Mason, I. C., Lattie, E. G., & Manalo, N. C. (2018). Feeling validated yet? A scoping review of the use of consumer-targeted wearable and mobile technology to measure and improve sleep. *Sleep Medicine Reviews*, 40, 151–159. <https://doi.org/10.1016/j.smr.2017.12.002>
- Baumert, M., Cowie, M. R., Redline, S., Mehra, R., Arzt, M., Pépin, J. L., & Linz, D. (2022). Sleep characterization with smart wearable devices: a call for standardization and consensus recommendations. *Sleep*, 45(12), Article zsac183. <https://doi.org/10.1093/sleep/zsac183>
- Berry, R. B., Quan, S. F., Abreu, A. R., Bibbs, M. L., DelRosso, L., Harding, S. M., Mao, M.-M., Plante, D. T., Pressman, M. R., Troester, M. M., & Vaughn, B. V. (2020). *The AASM manual for the scoring of sleep and associated events: Rules, terminology and technical specifications* (ed. 2.6). American Academy of Sleep Medicine.
- Bianchi, M. T. (2018). Sleep devices: wearables and nearables, informational and interventional, consumer and clinical. *Metabolism: Clinical and Experimental*, 84, 99–108. <https://doi.org/10.1016/j.metabol.2017.10.008>
- Birrer, V., Elgendi, M., Lambercy, O., & Menon, C. (2024). Evaluating reliability in wearable devices for sleep staging. *npj Digital Medicine*, 7(74), 1–14. <https://doi.org/10.1038/s41746-024-01016-9>
- Chapman, B. (2026, May 12). The meteoric rise-and problematic future-of wearable sleep tech. *Forbes*. Retrieved May 15 2026 from <https://www.forbes.com/sites/forbes-personal-shopper/2026/05/12/the-rise-and-future-of-wearable-sleep-tech/>
- Chinoy, E. D., Cuellar, J. A., Jameson, J. T., & Markwald, R. R. (2022). Performance of four commercial wearable sleep-tracking devices tested under unrestricted conditions at home in healthy young adults. *Nature and Science of Sleep*, 14, 493–516. <https://doi.org/10.2147/NSS.S348795>
- Cho, S., Ensari, I., Weng, C., Kahn, M. G., & Natarajan, K. (2021). Factors affecting the quality of person-generated wearable device data and associated challenges: Rapid systematic review. *JMIR Mhealth and Uhealth*, 9(3), Article e20738. <https://doi.org/10.2196/20738>
- Clawson, J., Pater, J. A., Miller, A. D., Mynatt, E. D., & Mamykina, L. (2015). No longer wearing: Investigating the abandonment of personal health-tracking technologies on Craigslist. In K. Mase, M. Langheinrich, & D. Gatica-Perez (Eds.), *UbiComp'15 - Proceedings of the 2015 ACM International Joint*

- Conference on pervasive and ubiquitous computing*, 647–658. Association for Computing Machinery. <https://doi.org/10.1145/2750858.2807554>
- de Zambotti, M., Goldstein, C., Cook, J., Menghini, L., Altini, M., Cheng, P., & Robillard, R. (2024). State of the science and recommendations for using wearable technology in sleep and circadian research. *Sleep*, 47(4), Article zsad325. <https://doi.org/10.1093/sleep/zsad325>
- de Zambotti, M., Menghini, L., Grandner, M. A., Redline, S., Zhang, Y., Wallace, M. L., & Buxton, O. M. (2022). Rigorous performance evaluation (previously, “validation”) for informed use of new technologies for sleep health measurement. *Sleep Health*, 8(3), 263–269. <https://doi.org/10.1016/j.sleh.2022.02.006>
- Dion, K., Porteous, M., Kendzerska, T., Nixon, A., Lee, E., de Zambotti, M., Garland, S. N., Singh, M., De Luca, G., Gillman, S., Baril, A. A., Gallson, D., Robillard, R., & Canadian Sleep Research Consortium (2025). Sleep, health care-seeking behaviors, and perceptions associated with the use of sleep wearables in Canada: Results from a nationally representative survey. *Journal of Medical Internet Research*, 27, Article e68816. <https://doi.org/10.2196/68816>
- Djanian, S., Bruun, A., & Nielsen, T. D. (2022). Sleep classification using consumer sleep technologies and AI: A review of the current landscape, *Sleep Medicine*, 100, 390–403. <https://doi.org/10.1016/j.sleep.2022.09.004>
- Doherty, C., Baldwin, M., Keogh, A., Caulfield, B., & Argent, R. (2024). Keeping pace with wearables: A living umbrella review of systematic reviews evaluating the accuracy of consumer wearable technologies in health measurement. *Sports Medicine*, 54, 2907–2926. <https://doi.org/10.1007/s40279-024-02077-2>
- Dong, M. (2025). Investigating the tracking anxiety and dependence of smartwatch users based on physiological and fitness data. *International Journal of Human-Computer Interaction*, 41(13), 8153–8166. <https://doi.org/10.1080/10447318.2024.2405785>
- Ghorbani, S., Golkashani, H. A., Chee, N. I. Y. N., Teo, T. B., Dicom, A. R., Yilmaz, G., Leong, R. L. F., Ong, J. L., & Chee, M. W. L. (2022). Multi-night at-home evaluation of improved sleep detection and classification with a memory-enhanced consumer sleep tracker. *Nature and Science of Sleep*, 14, 645–660. <https://doi.org/10.2147/NSS.S359789>
- Godfrey, A., Hetherington, V., Shum, H., Bonato, P., Lovell, N. H., & Stuart, S. (2018). From A to Z: Wearable technology explained. *Maturitas*, 113, 40–47. <https://doi.org/10.1016/j.maturitas.2018.04.012>
- Grandner, M. A., Bromberg, Z., Hadley, A., Morrell, Z., Graf, A., Hutchison, S., & Freckleton, D. (2023). Performance of a multisensor smart ring to evaluate sleep: in-lab and home-based evaluation of generalized and personalized algorithms. *Sleep*, 46(1), Article zsac152. <https://doi.org/10.1093/sleep/zsac152>
- Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., & Pedreschi, D. (2018). A survey of methods for explaining black box models. *ACM Computing Surveys*, 51(5), Article 93. <https://doi.org/10.1145/3236009>

- Guillodo, E., Lemey, C., Simonnet, M., Walter, M., Baca-García, E., Masetti, V., Moga, S., Larsen, M., HUGOPSY Network, Ropars, J., & Berrouiguet, S. (2020). Clinical applications of mobile health wearable-based sleep monitoring: Systematic review. *JMIR Mhealth and Uhealth*, 8(4), Article e10733. <https://doi.org/10.2196/10733>
- Imtiaz, S. A. (2021). A systematic review of sensing technologies for wearable sleep staging. *Sensors*, 21(5), Article 1562. <https://doi.org/10.3390/s21051562>
- Kainec, K. A., Caccavaro, J., Barnes, M., Hoff, C., Berlin, A., & Spencer, R. M. C. (2024). Evaluating accuracy in five commercial sleep-tracking devices compared to research-grade actigraphy and polysomnography. *Sensors*, 24(2), Article 635. <https://doi.org/10.3390/s24020635>
- Kazemi, K., Abiri, A., Zhou, Y., Rahmani, A., Khayat, R. N., Liljeberg, P., & Khine, M. (2024). Improved sleep stage predictions by deep learning of photoplethysmogram and respiration patterns. *Computers in Biology and Medicine*, 179, Article 108679. <https://doi.org/10.1016/j.combiomed.2024.108679>
- Klier, K., & Wagner, M. (2022). Agreement of sleep measures—a comparison between a sleep diary and three consumer wearable devices. *Sensors*, 22(16), Article 6189. <https://doi.org/10.3390/s22166189>
- Kuosmanen, E., Visuri, A., Kheirinejad, S., van Berkel, N., Koskimäki, H., Ferreira, D., & Hosio, S. (2022). How does sleep tracking influence your life? Experiences from a longitudinal field study with a wearable ring. *Proceedings of the ACM on Human-Computer Interaction*, 6(MHCI), Article 185. <https://doi.org/10.1145/3546720>
- Lee, T., Cho, Y., Cha, K. S., Jung, J., Cho, J., Kim, H., Kim, D., Hong, J., Lee, D., Keum, M., Kushida, C. A., Yoon, I.-Y., & Kim, J.-W. (2023). Accuracy of 11 wearable, nearable, and airable consumer sleep trackers: Prospective multicenter validation study. *JMIR Mhealth and Uhealth*, 11, Article e50983. <https://doi.org/10.2196/50983>
- Lee, Y. J., Lee, J. Y., Cho, J. H., Kang, Y. J., & Choi, J. H. (2025). Performance of consumer wrist-worn sleep tracking devices compared to polysomnography: A meta-analysis. *Journal of Clinical Sleep Medicine*, 21(3), 573–582. <https://doi.org/10.5664/jcsm.11460>
- Leota, J., Messman, B. A., Le, F., Jasinski, S., Capodilupo, E. R., Facer-Childs, E. R., & Wiley, J. F. (2026). How many nights are needed? The short-term stability of intraindividual variability in sleep parameters derived from accelerometry data in a cohort of normal sleepers. *Sleep*, Article zsag040. <https://doi.org/10.1093/sleep/zsag040>
- Lujan, M. R., Perez-Pozuelo, I., & Grandner, M. A. (2021). Past, present, and future of multisensory wearable technology to monitor sleep and circadian rhythms. *Frontiers in Digital Health*, 3, Article 721919. <https://doi.org/10.3389/fdgth.2021.721919>
- Mahir, A., Luong, N., Baryshnikov, I., Martikkala, A., Isometsä, E., & Aledavood, T. (2025). Multimodal sleep measurement and alignment analysis in outpatients with major depressive episode: Observational study. *JMIR Mhealth and Uhealth*, 13, Article e82465. <https://doi.org/10.2196/82465>

- Mathunjwa, B. M., Kor, R. Y. J., Ngarnkuekool, W., & Hsu, Y. L. (2025). A comprehensive review of home sleep monitoring technologies: Smartphone apps, smartwatches, and smart mattresses. *Sensors*, 25(6), Article 1771. <https://doi.org/10.3390/s25061771>
- Menghini, L., Cellini, N., Goldstone, A., Baker, F. C., & de Zambotti, M. (2021). A standardized framework for testing the performance of sleep-tracking technology: Step-by-step guidelines and open-source code. *Sleep*, 44(2), Article zsaa170. <https://doi.org/10.1093/sleep/zsaa170>
- Mogavero, M. P., Lanza, G., Bruni, O., Ferini-Strambi, L., Silvani, A., Faraguna, U., & Ferri, R. (2025). Beyond the sleep lab: A narrative review of wearable sleep monitoring. *Bioengineering*, 12(11), Article 1191. <https://doi.org/10.3390/bioengineering12111191>
- Nelson, E. C., Sools, A. M., Vollenbroek-Hutten, M. M. R., Verhagen, T., & Noordzij, M. L. (2020). Embodiment of wearable technology: Qualitative longitudinal study. *JMIR Mhealth and Uhealth*, 8(11), Article e16973. <https://doi.org/10.2196/16973>
- Óskarsdóttir, M., Islind, A. S., August, E., Arnardóttir, E. S., Patou, F., & Maier, A. M. (2022). Importance of getting enough sleep and daily activity data to assess variability: Longitudinal observational study. *JMIR Formative Research*, 6(2), Article e31807. <https://doi.org/10.2196/31807>
- Oura. (n.d.). *Oura Device Firmware Versions*. Retrieved 29 March 2026, from <https://support.ouraring.com/hc/en-us/articles/34036777934227-Oura-Device-Firmware-Versions>
- Pan, Q., Brulin, D., & Campo, E. (2020). Current status and future challenges of sleep monitoring systems: Systematic review. *JMIR Biomedical Engineering*, 5(1), Article e20921. <https://doi.org/10.2196/20921>
- Perez-Pozuelo, I., Zhai, B., Palotti, J., Mall, R., Aupetit, M., Garcia-Gomez, J. M., Taheri, S., Guan, Y., & Fernandez-Luque, L. (2020). The future of sleep health: a data-driven revolution in sleep science and medicine. *npj Digital Medicine*, 3(1), Article 42. <https://doi.org/10.1038/s41746-020-0244-4>
- Rentz, L. E., Ulman, H. K., & Galster, S. M. (2021). Deconstructing commercial wearable technology: Contributions toward accurate and free-living monitoring of sleep. *Sensors*, 21(15), Article 5071. <https://doi.org/10.3390/s21155071>
- Roos, L. G., & Slavich, G. M. (2023). Wearable technologies for health research: Opportunities, limitations, and practical and conceptual considerations. *Brain, Behavior, and Immunity*, 113, 444–452. <https://doi.org/10.1016/j.bbi.2023.08.008>
- Ryals, S., Chiang, A., Schutte-Rodin, S., Chandrakantan, A., Verma, N., Holfinger, S., Abbasi-Feinberg, F., Bandyopadhyay, A., Baron, K., Bhargava, S., He, K., Kern, J., Miller, J., Patel, R., Ratnasoma, D., & Deak, M. C. (2023). Photoplethysmography—new applications for an old technology: a sleep technology review. *Journal of Clinical Sleep Medicine*, 19(1), 189–195. <https://doi.org/10.5664/jcsm.10300>
- Schwartz, L. P., Devine, J. K., Choynowski, J., & Hursh, S. R. (2024). Consumer preferences for sleep-tracking wearables: The role of scientific evaluation and

- endorsement. *Sleep Health*, 10(2), 163–170.
<https://doi.org/10.1016/j.sleh.2023.11.009>
- Seçkin, A. Ç., Ateş, B., & Seçkin, M. (2023). Review on wearable technology in sports: Concepts, challenges and opportunities. *Applied Sciences*, 13(18), Article 10399. <https://doi.org/10.3390/app131810399>
- Tuovinen, L., & Smeaton, A. F. (2019). Unlocking the black box of wearable intelligence: Ethical considerations and social impact. *2019 IEEE Congress on Evolutionary Computation (CEC)*, 3235–3243. IEEE.
<https://doi.org/10.1109/CEC.2019.8790173>
- Van Gorp, H., Huijben, I. A. M., Fonseca, P., Van Sloun, R. J. G., Overeem, S., & Van Gilst, M. M. (2022). Certainty about uncertainty in sleep staging: A theoretical framework. *Sleep*, 45(8), Article zsac134.
<https://doi.org/10.1093/sleep/zsac134>
- Younes, M., Gerardy, B., Pack, A. I., Kuna, S. T., Castro-Diehl, C., & Redline, S. (2022). Sleep architecture based on sleep depth and propensity: patterns in different demographics and sleep disorders and association with health outcomes. *Sleep*, 45(6), Article zsac059. <https://doi.org/10.1093/sleep/zsac059>
- Zapalac, K., Miller, M., Champagne, F. A., Schnyer, D. M., & Baird, B. (2024). The effects of physical activity on sleep architecture and mood in naturalistic environments. *Scientific Reports*, 14(1), Article 5637.
<https://doi.org/10.1038/s41598-024-56332-7>